

Announcements

- Refresher topics posted
- Github repos provisioned
 - Claim yours now
- Reading added
 - Bolker bestiary of distributions

Today

- Random number streams
- Modeling workflow
- Bernoulli distribution for binary events
- Agent-based models
- Binomial distribution as emergent process
- Using modern RNG libraries
- Events in continuous time
- Exponential distribution

Modeling workflow: bio model

- Think about **biological processes** at the level of interacting "units" or "**agents**"
 - population ecology: individuals
 - physiology: cells or organelles
 - biochemistry: molecules
 - ecosystem ecology: individuals & molecules

Modeling workflow: bio model

- Think about **biological processes** at the level of interacting "units" or "**agents**"
- Make stochastic models for **small pieces**
 - stochastic model for death due to getting eaten by a predator
 - stochastic model for death due to competition or conflict with others
 - stochastic model for death due to other random events
 - stochastic model for birth
 - stochastic model for dispersal

Modeling workflow: bio model

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- **Simulate** the pieces to confirm intuition and correctness
 - check against known results ("unit tests")

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- **Simulate** the system
- Derive higher-level simplified models by **scaling up** check scaled-up model against ABM

Random number streams

- Keep track of the state of the RNG
- Feed back in to the iterative algorithm
- Continue using the same stream
- Each successive number is deterministic
 - repeatable, reproducible
- No need to set a new seed each time we want the next set of random numbers

Binary stochastic events

- Conceptualize a **biological process** as an **event** that occurs with **probability p**
 - death, disturbance, mutation, move ...
- Two outcomes (binary)
 - event occurs
 - event doesn't occur (or different event)
- Probability **represents uncertainty** about which outcome occurs

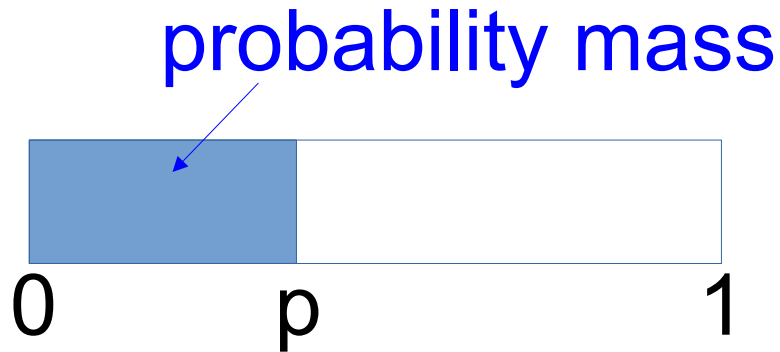
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This is the abstraction step: a model for the stochastic process

These assumptions $\longrightarrow$ **Bernoulli** distribution

Bernoulli RNG algorithm



Builds on uniform
random numbers

Algorithm

generate a random number U from $U(0,1)$

if $U \leq p$

 event occurs

else

 event doesn't occur (or other event occurs)

Bernoulli distribution

- Probability of a binary event

- $\Pr(X=1) = p$

Random variable
(our abstract concept)

- $\Pr(X=0) = 1 - p$

$$f(x; p) = \begin{cases} p & \text{if } x=1 \\ 1-p & \text{if } x=0 \end{cases}$$

Probability
mass
function
(pmf)

$$X \sim \text{Bernoulli}(p)$$

Agent-based models

- Biological processes from the perspective of **agents**
 - typically individuals in ecology/evolution
 - ... hence “individual based model” IBM
 - species, genes, nucleotide locations, molecules

Agent-based models

- Biological processes from the perspective of **agents**
- Processes are carried out or experienced by agents
 - predation: one agent eats another
 - death: the agent dies
 - birth: one agent produces new agents
 - dispersal: an agent moves from one location to another

Agent-based models

- Biological processes from the perspective of **agents**
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- **Explicitly track** each agent/individual

Agent-based models

- Biological processes from the perspective of **agents**
- Processes are carried out or experienced by agents
- **Explicitly track** each agent/individual
- Agents/individuals have **attributes/properties/traits**
 - alive/dead
 - location
 - age
 - size

Modern algorithms

- 1 generate random bits
- 2 turn those into integers
- 3 divide by largest integer to give $\text{Uniform}(0,1)$

Modern algorithms

- **R:** Mersenne_Twister
- **Numpy:** PCG64 (Permuted Congruential Generator)